

Modeling and control of fluid dispensing processes: a state-of-the-art review

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Abstract Fluid dispensing is a method by which fluid materials, such as epoxy, adhesive, and encapsulant, are delivered in a controlled manner in electronics packaging. This paper presents a brief review of past and recent developments in the modeling and control of the time-pressure fluid dispensing process. In particular, the characterization of the fluid flow behavior is addressed by reviewing several promising models from both time-independent and time-dependent perspectives. In the modeling of the time-pressure fluid dispensing process, various approaches for representing the flow rate of fluid dispensed and the profile of fluid formed on target are examined; and the issues involved are identified. In the control of time-pressure dispensing process, a brief review of various control methods is presented along with their limitations. The challenges associated with this control problem are also discussed. This paper is concluded with the recommendations of research in the future.

Keywords Time-pressure dispensing · Flow behavior · Modeling · Control

1 Introduction

Fluid dispensing is a method by which fluid materials (such as epoxy, adhesive, encapsulant) are delivered in a controlled manner. It has been widely used in various packaging processes in the electronics manufacturing

industry, such as advanced integrated circuit encapsulation (AICE) and surface mount technology (SMT). In AICE, one of the typical processes is die encapsulation, in which a die and the wires bonding between the die and a substrate need to be enclosed in an encapsulant in order to protect the die and wires from any environmental attacks [1]. Another typical process in AICE is the underfill encapsulation where, in order to reduce the thermal stress resulting from the difference in thermal-expansion coefficient between the flip chip and the substrate, the gap between them must be filled with an encapsulant [2, 3]. In these processes, the encapsulant is usually applied by means of dispensing. In SMT, dispensing is used to apply solder paste onto the pads of a printed circuit board (PCB) prior to the reflow-soldering process and/or to deliver adhesive onto the PCB in order to secure surface mount components in place during the wave-soldering process [4]. To achieve a package with high quality in the above applications, besides the proper selection of material, the amount of material dispensed and the profile of material formed on the substrate or PCB must be well controlled.

Figure 1 shows the schematic diagram of a time-pressure dispensing system typically used in electronic packaging. An air supply is used to provide pressurized air in combination with a valve to control the duration of pressurized air. Through a transmission line, pressurized air is applied to a syringe. Under the action of pressurized air, the fluid in the syringe is forced through the needle. Once extruded from the needle, the fluid drops onto a board, and then flows or spreads on the board until an equilibrium profile is formed. Time-pressure dispensing is a complex process, involving the air flow in both the transmission line and the syringe, the fluid flow in both the syringe and the needle, and the spreading of fluid on the board. Consequently, air compressibility and fluid flow

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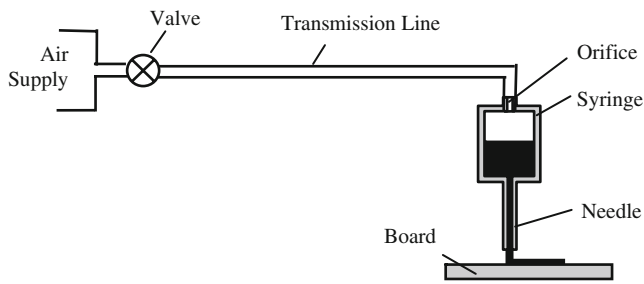


Fig. 1 Schematic of a time-pressure dispensing system typically used in electronic packaging

behavior are involved in the process and affect the process performance. For dispensing small amount of fluid, for example, the influence of air compressibility on the amount of fluid dispensed is particularly significant. Moreover, due to the fact the air volume in the syringe increases continuously with the process proceeding, the influence of air compressibility causes the inconsistency in amount dispensed under the identical operational conditions. It was observed by Dixon et al. [5] that the amount of the fluid dispensed could be reduced by 30% as the air volume in syringe varies from nil to full of the syringe. This is one of serious problems in the dispensing process [5–8]. Another problem observed in the dispensing process is the variation in the flow rate of fluid dispensed with time. This is mainly due to the fact that the flow behavior of the fluid being dispensed is time dependent [9–12].

To alleviate these problems, modeling and control of time-pressure dispensing has been drawing much attention from researchers in the past two decades; and various methods have been developed and evaluated with varying degrees of success. In this paper, a brief review of these methods is presented along with the key issues identified in this research area. Some recommendations are also presented for the future research.

2 Modeling of dispensing process

2.1 Characterization of flow behavior

For modeling of the time-pressure dispensing process, one of the key issues is to understand and characterize the flow behavior of the fluid being dispensed as it can significantly affect the process performance. The fluids employed in electronics packaging such as adhesives, epoxy, and encapsulant are usually non-Newtonian ones. One of the characteristics that distinguish non-Newtonian fluids from Newtonian fluids is the non-linear relationship between shear stress and shear rate at a given temperature. For some non-Newtonian fluids, such a relationship is also dependent on the time of shearing or the previous shear history [13].

In such cases, the flow behavior is said to be time-dependent, otherwise time-independent.

As far as the time-independent flow behavior is concerned, the most important characteristics are the non-linear relationship between shear stress and shear rate as well as its temperature dependency. There are various empirical models available in the literature [14]. It is noted that most of them were developed for liquid food products such as yoghurt, fruit juices, and milk. For characterizing the behavior of the fluids used in electronics packaging, there should be another consideration, i.e., the models used should be easily integrated in the development of models for the fluid dispensing process. One of such models is the Generalized Power Law equation, which is given by

$$\tau = \tau_0 + K \dot{\gamma}^n \quad (1)$$

where τ and τ_0 are the shear stress and the yield stress, respectively; $\dot{\gamma}$ is the shear rate; K is the consistency index with a unit of $\text{Pa}\cdot\text{s}^n$, and n is the flow behavior index (dimensionless). Razban and Davies [15, 16] used this model to characterize the flow behavior of Elastisol M23 (supplied by Evode Ltd.), in which $\tau_0=0$; and Han and Wang [2] used this model to investigate the flow behavior of Hysol FP4510 (supplied by Dexter Corporation, USA). In these studies, the values of n , τ_0 , and K were determined by using rheometers. Specifically, for a given fluid, n was usually identified as a constant; while τ_0 and K as functions of temperature to account for the temperature influence.

Due to the fact that the flow behavior is sensitive to temperature, the application of the flow behavior characterized by means of a rheometer requires that the temperature in the dispensing process be controlled the same as the one in the rheometer. Otherwise, any difference can result in a large error in the model predictions by using the characterized flow behavior. To alleviate this problem, one of the methods is to characterize the flow properties directly from the dispensing process [17], rather than a rheometer. In this method, the model developed for the dispensing process is characterized by a non-linear function and used to map the process parameters to the process performance, depending on the values of the parameters associated with the characterization of the flow behavior, i.e., n , τ_0 , and K in Eq. (1); and then, based on the data of both known process parameters and measured process performance, the values of n , τ_0 , and K are determined by using non-linear least-squares data fitting. This is a cost- and time-effective way to characterize the flow behavior from the dispensing process. However, the so-characterized flow behavior may not be applicable to other systems due to the possible temperature difference between the dispensing process and the systems applied to.

The characterization of time-dependent flow behavior is much more complicated than that of the time-independent

one. It is known that time-dependent fluids can be either thixotropic if the viscosity (i.e., the ratio of shear stress to shear rate) decreases with time or rheopectic if the viscosity increases with time [13]. To the best of author's knowledge, most of the fluids in electronics packaging exhibit thixotropic behavior. One model given in literature for the thixotropic behavior is the time-dependent Generalized Power Law equation. This model was proposed by Matinex–Paradilla and Hardy [18] based on the Generalized Power Law equation, taking the form of

$$\tau = \tau_0 + K \dot{\gamma}^n t^{-m} \quad (2)$$

where t is the shearing time; m is the thixotropic index (dimensionless). Physically, m is a measure of the degree of time dependency—the greater departure of m from zero, the more pronounced the time dependency. Equation (2) has been utilized with various degrees of success for the characterization of time-dependent behavior of liquid food products. However, there are two drawbacks of this equation. One is that the equation is not applicable to describe the equilibrium state of viscosity because, as time becomes very large, the second term on the right side of Eq. (2) reduces to zero and the fluid viscosity is only dependent upon the yield stress, which is obviously not true. The other drawback is that this equation is not valid for the characterization of the recovery of viscosity, i.e., the viscosity gradually increases with time upon the removal of shearing, which always happens in the dispensing process due to the interruption of process. Therefore, models that can overcome the above drawbacks are desired.

One such model is based on structural theory. Cheng and Evans [19] first proposed structural theory to describe the time-dependent flow behavior. Since then, the theory has been greatly extended by other researchers. The basic idea of structural theory is the use of a structural parameter, λ , to account for the time-dependent effects on flow behavior. This parameter is associated with the internal structure of a fluid, which will break down under shearing and recover upon the removal of shearing. The fluid viscosity is then assumed to associate with λ . Two equations are usually used in structural theory: one is the state equation to describe the state of shear stress; and the other one is the rate equation to describe the rate of change in structural parameter, which are given by

$$\tau = \eta(\dot{\gamma}, \lambda) \cdot \dot{\gamma} \quad (3)$$

$$\frac{d\lambda}{dt} = g(\dot{\gamma}, \lambda) \quad (4)$$

where $\eta(\dot{\gamma}, \lambda)$ is the viscosity and $g(\dot{\gamma}, \lambda)$ is the rate of change in the structural parameter, both of which are dependent on the shear rate and the structural parameter. In

the literature [20–22], there are various forms of Eqs. (3) and (4), which were developed based on the flow behavior of the fluid concerned.

The characterization of flow behavior is a challenging task because the behavior may depend not only on temperature but also on the shearing time. There is no a universal equation applicable to all fluids. For a given fluid, the key issue is to select a suitable model based on the recorded flow behavior. Meanwhile, the model selected should also be easily integrated in the development of models for the fluid dispensing process, which is discussed in the next section.

2.2 The flow rate of fluid dispensed

The flow rate of fluid dispensed is one of the important performance indexes in the dispensing process. It is shown by experiments that the flow rate can be manipulated or affected by a number of factors such as the process parameters (e.g., the air pressure applied to the syringe and the system temperature), the structural parameters (e.g., the needle diameter and length), and the flow behavior of the fluid being dispensed. To represent the flow rate, one of the methods is to develop empirical models based on experimental data. Such a model, for example, was developed by using fuzzy regression approach [23]. It is noted in the development of empirical model that massive experiments are always required, which has proven to be time-consuming. As a result, developing analytical models for the flow rate of fluid dispensed, based on physical laws or principles, has been drawn much more attention from the researchers.

One modeling method is treating the flow in the needle as a fully developed laminar flow of Newtonian fluid. Thus, the well-known Poiseuille equation can be used to establish the flow rate. In the other hand, the assumption of Newtonian fluid greatly limits its applications. Another method is considering the fluid as non-Newtonian. Under the assumptions that: (1) the fluid is incompressible and time-independent, (2) the fluid is in steady laminar flow, (3) there is no slip between the fluid and the needle wall, and (4) the relationship between shear stress and shear rate of the fluid is defined by $\dot{\gamma} = f(\tau)$, the flow rate, Q , through the needle can be given by [13]

$$Q = \int_0^R u(r) 2\pi r dr = \frac{\pi R^3}{\tau_w^3} \int_0^{\tau_w} \tau^2 f(\tau) d\tau \quad (5)$$

in which $u(r)$ is the local velocity at the radius, r , where $r \leq R$ (i.e., the radius of the needle); τ and τ_w are the shear stresses at r and R , respectively, which are dependent on the frictional pressure drop in the needle and given by

$$\tau = \frac{r \Delta P}{2L}; \text{ and } \tau_w = \frac{R \Delta P}{2L} \quad (6)$$

where L is the needle length. If the relationship between shear stress and shear rate is described by the Generalized

Power Law Eq. (1), the flow rate of the fluid through the needle can be derived and given by [24]

$$Q = \frac{\pi R^3}{K^{1/n} \tau_w^3} (\tau_w - \tau_0)^{(n+1)/n} \times \left[\frac{n}{3n+1} \tau_w^2 + \frac{2n^2}{(2n+1)(3n+1)} \tau_w \tau_0 + \frac{2n^3}{(n+1)(2n+1)(3n+1)} \tau_0^2 \right] \quad (7)$$

The method described above has been successfully employed to investigate the flow rate in the fluid dispensing process [15, 16, 25]. It is noted in the above method that the flow rate is obtained under the assumption that the fluid is time-independent. As mentioned previously, many fluids in electronic packaging, however, exhibit time-dependent flow behavior. To account for the influence of time-dependent fluid behavior on the dispensing process performance, a straightforward method may be to incorporate the equations used to characterize the time-dependent flow behavior into the development of models for the dispensing process. The key issue here is that the shearing history that the fluid has experienced needs to be known and inputted to the model as the prior knowledge. This may not be an effective method because it is hard to describe the shearing history of fluid in a dispensing process. Along a different research avenue, Chen et al. [10] developed a method based on the technique of model updating. The basic idea of this method is that a model of the dispensing process is first developed under the assumption that the fluid is time-independent; and then, in order to account for the influence of time-dependent fluid behavior on the dispensing process, the parameters associated with fluid behavior, such as τ_0 , K , and n in Eq. (1), are considered as time-variable and updated as the dispensing process proceeds. In this study, it is shown that the model with the so-updated parameter can be used to represent the influence of time-dependent fluid behavior on the performance of the dispensing process, and that this method is appropriate to be integrated into the control scheme for improved dispensing performance, which is discussed in the following section.

Besides the time-dependent flow behavior, the influence of air compressibility is another important factor to affect the dispensing performance. Due to the compressibility of air in both transmission line and syringe, as shown in Fig. 1, the air pressure in the syringe is a dynamic response to the pressure of the air supply; and, as a result, the flow rate of fluid dispensed is also a dynamics response. Therefore, modeling the dynamics of flow rate is essentially required, especially in the applications such as SMT, where small amount of fluid needs to be dispensed.

In the literature, several dynamic models have been reported. Most of these models were developed based on system identification. One such model was developed by Razban [15]. Based on experiments, the dynamics of the

profile size of fluid formed on the board (which is proportional to the flow rate) to the voltage applied to the dispensing valve (which manipulates the air pressure) was identified as a second order system with the following transfer function

$$G(s) = \frac{322.13 K_p}{s^2 + 21.18s + 322.13} e^{-T_l s} \quad (8)$$

where K_p is the process gain and T_l is the lumped time delay. Both of them are dependent upon the operational conditions. Another empirical dynamic model is the auto regressive with external input (ARX) model, which was developed by West et al. [8]. It is given by

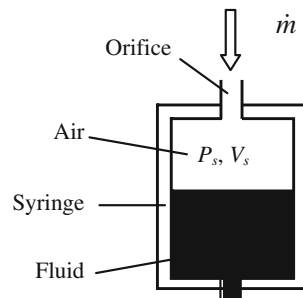
$$\begin{aligned} y(nT) + a_1 y(nT - T) + \dots + a_n y(0) \\ = b_0 P(nT) + b_1 P(nT - T) + \dots \\ + b_m P(nT - mT) + e(nT) \end{aligned} \quad (9)$$

where T is the sampling interval, y is the blob area of the adhesive dispensed (which is proportional to the amount dispensed), P is the air pressure, and e is the error term. Auto-regression techniques were employed to determine the coefficients: a_i ($i=1, 2, \dots, n$) and b_j ($j=1, 2, \dots, m$).

The above models did facilitate the design of process controllers because of the linear dynamics of the input–output relationship but could not provide information related to the influence on the flow rate of such parameters as the transmission-line length, syringe size, and needle sizes. Also, any change in the aforementioned parameters requires the development of a new model. Obviously, this strategy is very costly and may be difficult to implement in industry. Moreover, as discussed previously, one serious problem in the dispensing process is the inconsistency in the amount of fluid dispensed under the identical operational conditions. Using these empirical models, it is extremely difficult to gain insight into the problem. Thus, physical models representative of the flow rate dynamics are obviously desirable and advantageous.

Driven by the need, researchers also looked at the development of such physical models [7, 26], by taking the compressibility of air in the transmission line and syringe into account. It is noted in time-pressure dispensing that the syringe contains both air and fluid, with the top connected to the air supply and the bottom to a needle, as shown in Fig. 2. One distinct characteristic is that, as the dispensing

Fig. 2 Schematic of a syringe containing both air and fluid



proceeds, the volume of fluid in the syringe decreases while the air volume increases. This is considered the main reason for the inconsistency in the amount of fluid dispensed. In the study [7], the syringe was modeled as a variable volume under the assumption that the air pressure in the syringe was uniform. Thus, the air pressure in the syringe, P_s , is governed by [27]

$$\frac{dP_s}{dt} = -\frac{\nu P_s}{V_s} \frac{dV_s}{dt} + \frac{\nu \chi T_s}{V_s} \dot{m} \quad (10)$$

where V_s and T_s are the air volume and temperature of the syringe, respectively, $\dot{m}$ is the mass-flow rate of the air into the syringe through the orifice, ν the ratio of the specific heats (1.4 for air), χ a constant (287 J/kg/K for air). It can be seen from the above equation that the air pressure in the syringe (P_s) is dependent on the air volume (V_s). As a result, the fluid dispensed varies, depending on the air volume, under the identical pressure pulse of the air supply.

2.3 The profile of fluid formed on boards

Besides the flow rate of fluid dispensed, the fluid profile formed on the substrate or board is another important performance index. Depending on whether the needle carried by the dispensing head moves horizontally or not, two kinds of profiles are usually formed: a line and a dot, as schematically shown in Fig. 3. The cross-section sizes are measured by the width and height (denoted by W and H , respectively) for a line, and the diameter and height (denoted by D and H , respectively) for a dot.

To determine the cross-section size, it was assumed by Razban and Davies [16] that the line was cylindrical in shape by neglecting the spreading of fluid on a board. If the

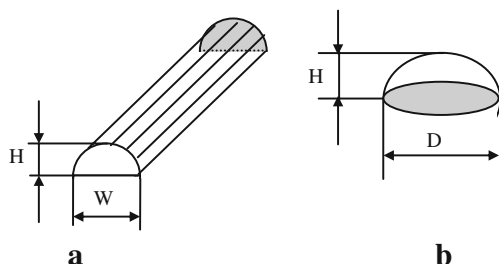


Fig. 3 Profiles of fluid **a** a line and **b** a dot

spreading process is taken into account, the surface tension i.e., the force exerted on the surface due to intermolecular interactions must be considered. By including the surface tension effect, Tummala and Rymaszewski [3] gave the height of an equilibrium dot as

$$H = \sqrt{\frac{2\sigma(1 - \cos \theta)}{\rho g}} \quad (11)$$

where σ is the surface tension, θ the angle at the contact line between the fluid and the board, ρ the fluid density, and g the gravitational acceleration.

In general, the spreading of fluid on a board is a fundamental problem in fluid mechanics, in which bulk, surface, and line forces are coupled through a free boundary. The configuration of this problem is shown in Fig. 4, in which fluid is surrounded by ambient air and spreading on a horizontal and flat board. The spreading is considered in either two-dimensional Cartesian coordinates (x, z) for a line or axisymmetric cylindrical coordinates (r, z) for a dot. In Fig. 4, a represents the contact line between the fluid and the board, which moves with the spreading of fluid on the board until an equilibrium state is reached; and h denotes the interface between the fluid and the ambient air. The velocity and thermal fields in the fluid are governed by the Navier–Stokes, continuity, and energy equations.

Given the boundary and initial conditions, one could solve the aforementioned equations. In general, however, it is very difficult as this is a moving-boundary problem for the Navier–Stokes equations. The problem can be simplified by assuming that the ratio of the height to the width of a line or the diameter of a dot is very small [28–30]. However, this assumption is not applicable for fluid dispensing because the height is usually in the same order as the width or the radius. Chen et al. [25] developed a method to determine the profile of fluid formed in the dispensing process for electronic packaging. The basic idea of the method is that, for electronic packaging, the concern is not the spreading process, but the outcome of this process, i.e., the final equilibrium profile of the fluid formed. Therefore, one can set up equilibrium conditions

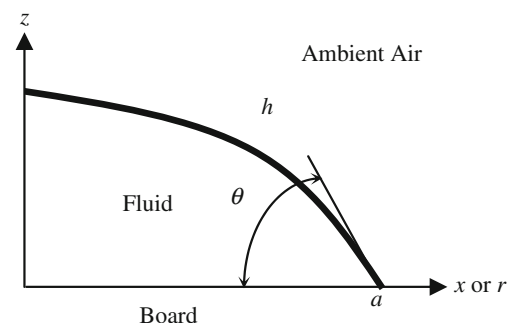


Fig. 4 Half schematic of the spreading of fluid on a board

by assuming that there is no movement in the fluid and that the fluid reaches the same temperature as the board. By applying these conditions, one can significantly simplify the governing equations. Based on the simplified equations, the numerical solution to the profile of the interface, h , was obtained by Chen et al. [25].

The models reviewed in this section are summarized in Table 1, along with their pros and cons. The key issues in developing models for the dispensing process are the representation of the influence of time-dependent flow behavior, which the fluid being dispensed may exhibit, and

the representation of the influence of the change in air volume in the syringe as the dispensing process proceeds. Addressing these issues in existing models leaves a lot to be desired, which is further discussed in Section 4.

3 Control of dispensing process

As discussed previously, there are two basic types of fluid dispensing for electronic packaging, i.e., dot and line. In the process of dispensing dots, the dot amount is of importance,

Table 1 Summary of the models for the fluid dispensing process

	Models	Pros and cons
Characterization of flow behavior	Generalized Power Law equation [2, 15–17]:	For time-independent flow behavior. Parameters identified by using either rheometers or dispensing systems. Easily integrated in the model development for the fluid dispensing process.
	Time-Dependent Generalized Power Law equation [18]:	For time-dependent flow behavior. Not applicable to the equilibrium state of viscosity. Not applicable to the viscosity recovery upon the removal of shearing.
	Structural Theory [19–22]:	Shearing history needed to be known. For time-dependent flow behavior. Various forms available. Complicated parameter identification. Shearing history needed to be known.
Representation of the flow rate of fluid dispensed.	Models for time-independent fluids [15, 16, 25].	Simple and approximate. For dispensing time-independent fluids. For the steady-state flow rate.
	Models based on model updating [10]	For dispensing time-dependent fluids. Not including the influence of air compressibility. Easily integrated in the control schemes for improved performance.
	Models based on system identification [8, 15]	For the dynamic flow rate. Including the influence of air compressibility. Not applicable to time-dependent fluids. Easily integrated in the control schemes for improved performance.
	Physical models with consideration of air compressibility [7, 26]	No information provided on the influence of structural parameters. For the dynamic flow rate. Including the influence of air compressibility. Not applicable for time-dependent fluids. Easily integrated in the control schemes for improved performance.
Representation of the profile of fluid formed on boards	Cylindrical shape [16]	With information on the influence of structural parameters.
	Tummala and Rymaszewski's model [3]	Not considering the spreading of fluid on the board. Simple and approximate. Considering the influence of surface tension.
	Chen's model [25]	For the equilibrium profile. Considering the influence of surface tension and contact angle. Not applicable to the fluid profile during spreading.

which is an integration of flow rate within the dispensing time period; and the control of dispensing process is a discrete problem. In the process of dispensing lines, the concern is the flow rate and the control is a continuous problem. In both cases, one of the fundamental requirements for electronic packaging is to maintain the consistency of the process performance [31]. It is observed that the process performance can be degraded by some unavoidable change in the dispensing process such as the time-dependent fluid behavior and the continuous increase of the air volume in the syringe. To improve the consistency, a number of control methods were developed and reported in the literature.

West et al. [8] investigated the control of dispensing dots. The background of their studies was dispensing adhesive by means of the testbed shown in Fig. 5. Through a voltage-controlled pressure regulator and solenoid valve, the height and width of the pressure pulse were manipulated to control the amount of adhesive dispensed. A vision feedback on the planar area of the dispensed dot was obtained using a charge-coupled-device (CCD) camera that was mounted on the dispensing head; and the measured planar area was used to determine the amount of fluid dispensed. In this study, system identification was used to develop the model reviewed in the preceding section; and, based on the developed model, an incremental proportional controller was applied to maintain the consistency in the amount dispensed. In addition, this study also showed that intelligent methods, such as knowledge-based, fuzzy logic, and neural network, could be included in the control system for fault detection, for example, the failure of pressure pulse.

One of the drawbacks of the above control system is the use of the camera vision, which not only complicates the control system and but also dramatically increases the cost of control. In addition, because the fluid is in flowing on the board, the measurements using a vision system may not

be stable and repeatable, thereby degrading the control performance. As a result, the method of off-line control instead of on-line is much preferred in industry. In off-line control, the measurements of the dispensed amount of fluid were made through breaking the regular dispensing routine and then dispensing samples of fluid. For the purpose of measurements, either a scale was used to weigh the dot amount [11] or a specially-developed apparatus was used to evaluate the dot size [6]. Once the dot amount or size was measured, the control action based on the incremental controller, or Bang-Bang controller was established to regulate the manipulating parameters to improve the consistency in the amount of fluid dispensed [6,11]. As discussed previously, the variation in the amounts of fluid dispensed mainly results from the increase in the air volume in syringe as the dispensing process proceeds. Consequently, in the above studies the measurements of the dispensed amount are critical and frequently required. Meanwhile, the improvement of consistency is limited by the use of simple Bang-Bang controllers.

Obviously, for such a control problem, the integration of a dynamics model, which can predict the variation in the amount of fluid dispensed, into control system would be greatly advantageous. For this purpose, Chen et al. [32] developed a control strategy, which is schematically shown in Fig. 6. The “actual” dot amount for feedback is predicted by a model, which is updated by using the air volume in the syringe. The predicted dot amount is compared to the desired one; and their error is fed to a compensator, which is used to manipulate the magnitude of the air pressure pulse for achieving the desired dot amount. This developed control strategy was validated experimentally by dispensing Newtonian fluids; and its application to dispensing non-Newtonian fluids that exhibit time-dependent flow behavior, however, remains to be addressed.

For the control of dispensing lines, several studies were conducted by Razban et al. [15, 16]. The background of their studies was dispensing adhesive or sealant onto workpieces in order to have structurally bonded joints in automotive and aerospace industries. A customer-made vision system including a CCD camera and a laser diode was used for measuring the line size. Based on system identification, an ARX model for the process dynamics was developed, which is reviewed in the preceding section. Based on the model, various conventional and predictive controllers were evaluated using either simulation or experimental studies. It was found that proportional (P) control was not applicable because of the presence of the steady-state error. For the proportional-integral (PI) control, the integral component has a considerable effect on forcing the steady-state error to be zero. However, due to the relatively long time delay in the dispensing process (typically 50–200 ms [15]), PI control did not work as

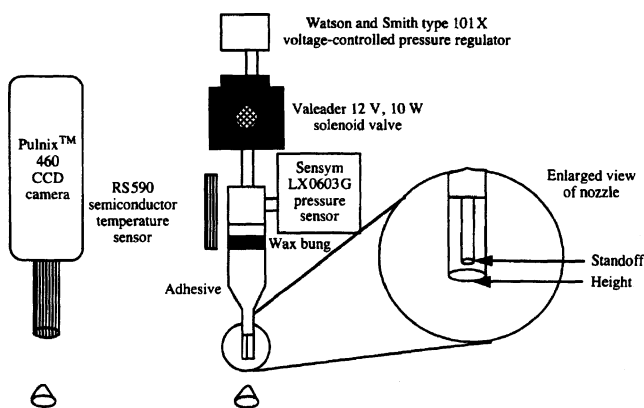
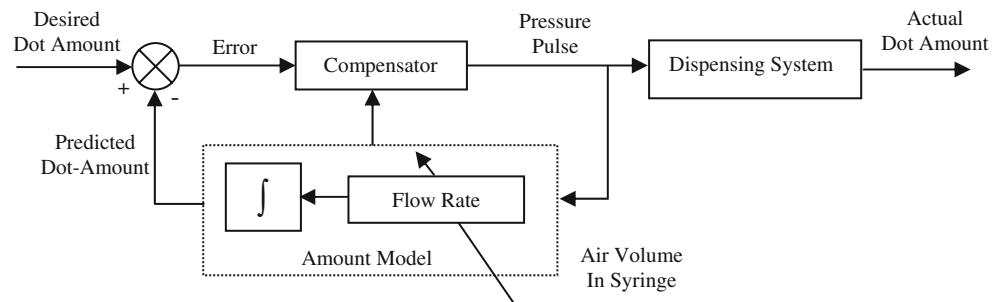


Fig. 5 Adhesive dispensing system [8]

Fig. 6 Control of dispensing dots [32]

effectively as it does in a process without a time delay. Predictive control was proven to be more effective in control of such a process. In the above various controllers, on-line measurements of the process performance by means of a vision system were used. As mentioned previously, on-line measurements may not accurate and repeatable due to the fact the fluid dispensed on a board is in spreading. Therefore, the improvement of the process performance is also limited.

Similar to dispensing dots, off-line control is also preferred in industry for dispensing lines. Bouras et al. [12] patented a technique for the dispensing process control, which is called mass-flow calibration. The idea of this technique is that the flow rate is measured periodically by weighing the fluid dispensed over a certain time interval for identifying the change in fluid viscosity. The air pressure is then adjusted based on a correction table as obtained experimentally such that the influence of the change in viscosity on the flow rate can be compensated. Of this technique, one limitation is assumption that the shear stress varies linearly with shear rate. This assumption is not appropriate for fluids used in electronic packaging. The other limitation is that the extent of the performance improvement is dependent on the accuracy and size of the correction table as applied to a range of dispensing conditions. Chen et al. [10] developed another off-line control strategy and successfully applied it to the control of dispensing lines on a typical commercial dispensing system (DS500, supplied by ASM, Hong Kong). In this study, the influence of variation in the flow rate of fluid dispensed due to the time-dependent fluid behavior was represented based on the techniques of model updating, as reviewed in the

preceding section; and an off-line control for compensating for this influence was then developed, which is shown schematically in Fig. 7. In this control strategy, the actual flow rate was measured off-line periodically and fed back to update the model for identifying the influence of the time-dependent fluid behavior; and then, based on the updated model, a compensator is used to regulate the manipulated parameters in order to achieve the desired flow rate.

The control methods reviewed in this section are summarized in Table 2, along with their pros and cons. In on-line control, the repeatability of the measurements by means of a cameral vision system is a concern due to the fact that the fluid is in spreading, thereby limiting the performance improvement. Thus, the off-line control, by eliminating the need of on-line measurements, is preferred in industry. In order to the maintain the consistency in the process performance, the compensation for the influence of the varying air volumes in the syringe and/or the influence of time-dependent flow behavior on the performance is the key issue in the control of the dispensing process.

4 Some recommendations of research for the future

With the dies, chips and components in electronics packaging becoming smaller and smaller, the demand for more precise fluid dispensing is increasing. To meet this need, there are some issues that remain to be addressed and clarified.

- (1) There is a need to develop a model representative of the amount of fluid dispensed, which is minute down to the order of microliters or nanoliters. Figure 8 shows

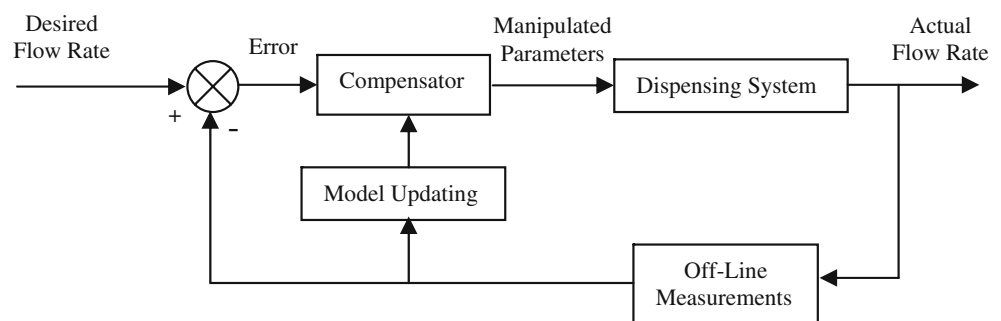
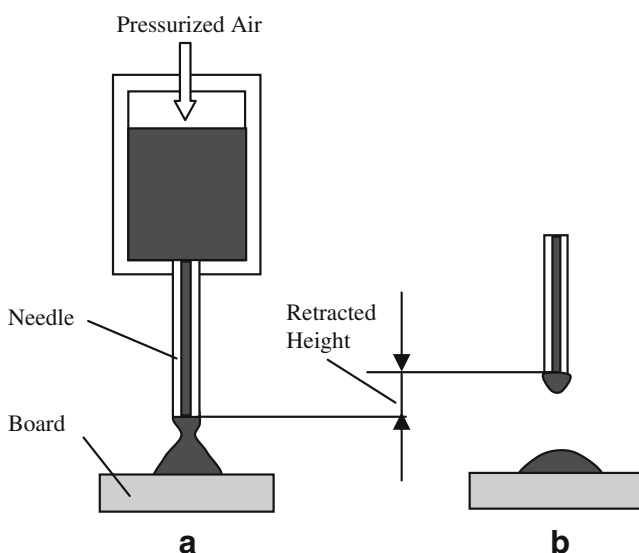
Fig. 7 Control of dispensing lines [10]

Table 2 Summary of control methods for the fluid dispensing process

	Controllers	Pros and cons
On-line control	P, PI, and predictive control [8, 15, 16]:	For dispensing dots and lines. The use of a camera vision system, causing the complexity in control and the increase in cost. Measurement repeatability is a concern as the fluid is in spreading.
	Model-based control [32]	For dispensing dots. Need a model. Compensate for the influence of the change in the syringe air volume. No measurements required
Off-line control	Bang-Bang control [11]:	For dispensing dots. Need to break the regular dispensing routine and dispensing samples for measurements.
	Control based on the mass-flow calibration [12]	For dispensing lines. For dispensing time-dependent fluids. Need to break the regular dispensing routine and dispensing samples for measurements.
	Control based on model updating [10]	Need a correction table. For dispensing lines. For dispensing time-dependent fluids. Need a model.
		Need to break the regular dispensing routine and dispensing samples for measurements.

schematically such a typical dispensing process. First, the dispensing head is transported to the dispensing position shown in Fig. 8(a). Under the action of air pressure, a minute volume of fluid is extruded through the needle, contacting with the board surface. As a result of surface tension, the fluid rapidly flows onto the surface and forms a bridge between the needle and the surface. The dispensing apparatus is then lifted straight up to the retracted position shown in Fig. 8(b).

**Fig. 8** Dispensing minute volume of fluid

Meanwhile, the fluid bridge is deformed into a necked configuration until it is broken. Given that the volume of fluid dispensed is minute and that the fluid dispensed usually exhibits compressibility and non-Newtonian behavior, the flow in both the syringe and the needle is transient compressible and viscous, and the dynamics of the flow rate is of particular interest in this application. At present, the dynamics was usually investigated under either the assumption that the compressibility is ignored or the assumption that the fluid is Newtonian. By releasing both the assumptions, the development of models to represent the dynamics of flow rate could be one of the research initiatives that should be undertaken in the future.

In addition, it is noted that in all the reported studies, the fluid volume flowing onto the target surface is assumed to be equal to the fluid volume extruded from the needle. In reality, however, this is not true. As the dispensing apparatus is lifted up and the fluid bridge is broken, one part of the volume extruded from the needle will flow onto the board and the other part will attach on the needle, as shown in Fig. 8. This is a problem associated with the fluid/liquid bridge in fluid mechanics, which has attracted considerable attention from researchers [33]. Application of existing knowledge into the dispensing process to represent the volume of fluid flowing onto the board would be another research initiative.

- (2) Developing a model to represent the influence of time-dependent flow behavior on the dispensing process would be another research initiative. As discussed previously, this task is demanding because the current state-of-the-art in fluid mechanics does not provide an effective method. One of methods is the use of model updating as reported by Chen et al. [10]. The issue of this method, which remains to address, is that the information available is less than the information required for model updating due to that fact that several parameters are needed to update and only one measurement data is available. To overcome this drawback, one of the solutions could be to measure some other process parameters, such as the pressure at the bottom in the syringe, which can be affected by the time-dependent flow behavior. As the result, the measured parameters could greatly enhance the inference information for the model updating.
- (3) It should be noted that the existing model-based control methods for dispensing process were developed to compensate for either the influence of the different air volumes in the syringe or the influence of time-dependent flow behavior on the dispensing performance. Develop a control method that can simultaneously compensate for both influences would be one of the research focuses in the future. To do that, the use of the adaptive control seems very promising to this particular control problem.

5 Conclusions

Time-pressure fluid dispensing is a complex process. Increasing demand for precision fluid dispensing in electronic packaging has raised the need for modeling and control of the dispensing process. This paper presents a review on various approaches reported in literature and identifies the issues in this research area, with the main conclusions highlighted as follows.

- (1) The flow rate of fluid dispensed is one of the important indexes to measure the performance of the dispensing process. For modeling of the steady-state flow rate in the time-pressure dispensing, the treatment of laminar flow of non-Newtonian fluids can be generally be adopted. For representing the dynamics of the flow rate, empirical dynamic models developed using the system identification may be used. The main drawback of these models is that they do not provide information related to the influence of the structural parameters. To overcome this drawback, physical dynamic models are essentially needed, in which air compressibility in both the transmission line and the

syringe must be taken into account. If the fluid dispensed exhibits time-dependent behavior, the flow rate of fluid dispensed can vary with time. In such a case, model updating proves to be an effective method to represent the variation in the flow rate. The development of dynamic models, by taking into account both air compressibility and time-dependent flow behavior, will be a research initiative worthwhile to pursue in the future.

- (2) The profile of the fluid formed on a target is another important index to measure the performance of the dispensing process. Generally, it is difficult to represent this profile because it results from the spreading process of fluid, which is governed by the Navier–Stokes, continuity, and energy equations with moving-boundary conditions. For electronic packaging, in which the equilibrium profile of fluid is concerned, one possible solution to this problem is to establish the equilibrium conditions of the spreading process and then to apply these conditions into the general governed equations for simplification.
- (3) One fundamental requirement for the control of fluid dispensing in electronic packaging is to maintain the consistency in the process performance (i.e., the amount for dispensing dots or the flow rate for dispensing lines). To fulfill this requirement, the typical practice is the use of a feed back control based on on-line measurements by means of a vision system. Due to the fact that the fluid dispensed on a board is in spreading, the measurements using the vision system may not be accurate and repeatable, thus limiting the improvement of the process performance. As a result, the use of off-line control is much preferred in industry. To do that, control based on model updating seems promising to this particular problem. In the future, the development of a control method to compensate for the influence of the varying air volumes in the syringe and the influence of time-dependent flow behavior on the dispensing performance simultaneously is one of the research initiatives.

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